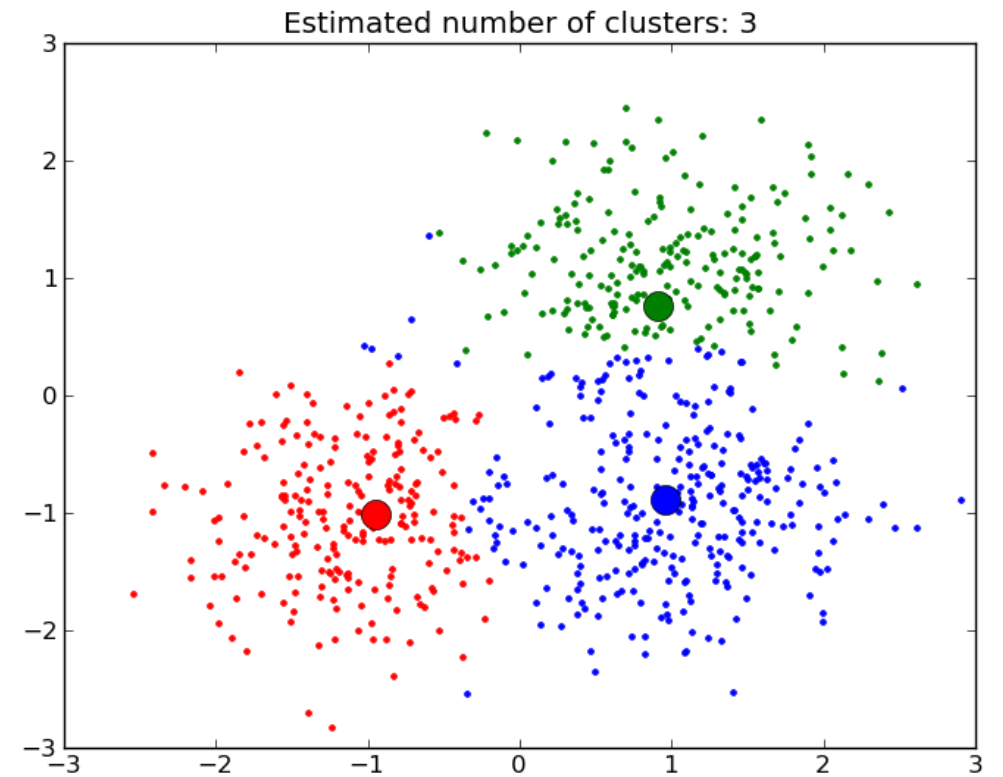


Clustering

Given a set of **examples** (points in some high-dimensional space)
find **groups** of similar elements



Clustering



Examples

- Find Amazon **users** sharing **similar** characteristics: for the **kind of objects** they buy, for the **budget** they exploit, for the **frequency** of orders they make
- Group Facebook **users** into communities because of their **friendships** and the **contents** they like
- Given a collection of **text documents**, find those regarding the **same topic**

Clustering



No supervisions implies...

- No **target** to be computed
- No straightforward **performance measure** to **evaluate** the success of our algorithm
- **Heuristic** approaches to assess validity of results
- Partial lack of **theoretical** grounds

Clustering

Main idea of clustering algorithms

- Partition data samples into K sets (named **clusters**)
 - All examples belong to one and only one cluster
- High **intra-cluster similarity**
- High **inter-cluster dissimilarity**
- Based on a **distance/metric** between examples

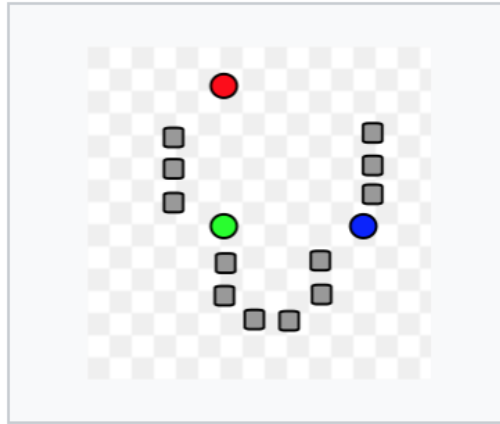
Clustering

The most famous clustering algorithm is **K-Means**:

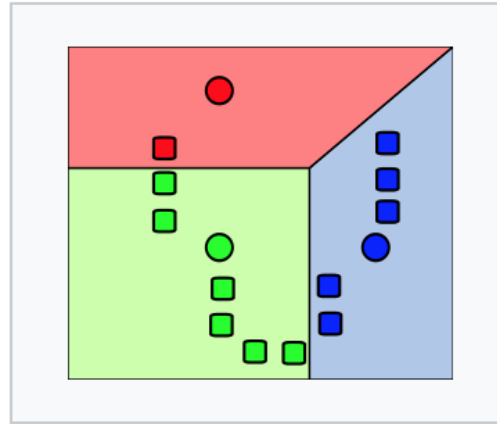
- Choose a number of clusters K (**in advance!!!**)
- Start with K **random** cluster centroids
- **Assign** each point to the **closest** centroid
- Compute **new** cluster centroids
- Iterate until **convergence** (?)

Clustering

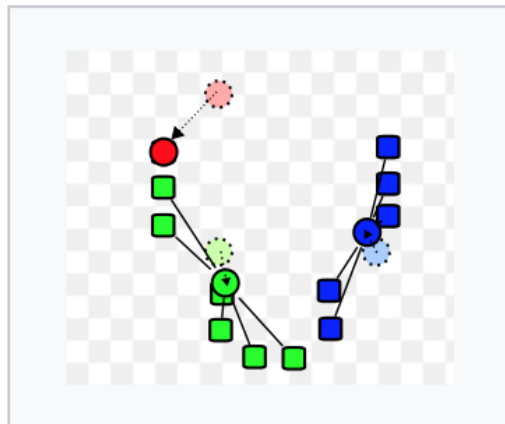
1. INIT



2. ASSIGN



3. COMPUTE



4. CONVERGE

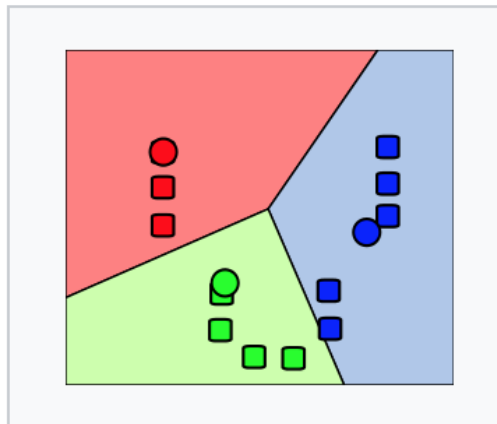


Image from Wikipedia

Clustering

K-Means algorithm

1. **init** centroids (K random points)

2. **repeat**

2a. **assignment** $s(x) = \arg \min_{c_i \in C} d(c_i, x)$

2b. **compute**
$$c_i = \frac{1}{|x, s(x) = c_i|} \sum_{x, s(x)=c_i} x$$

3. **until** stopping criterion

Clustering



Pros

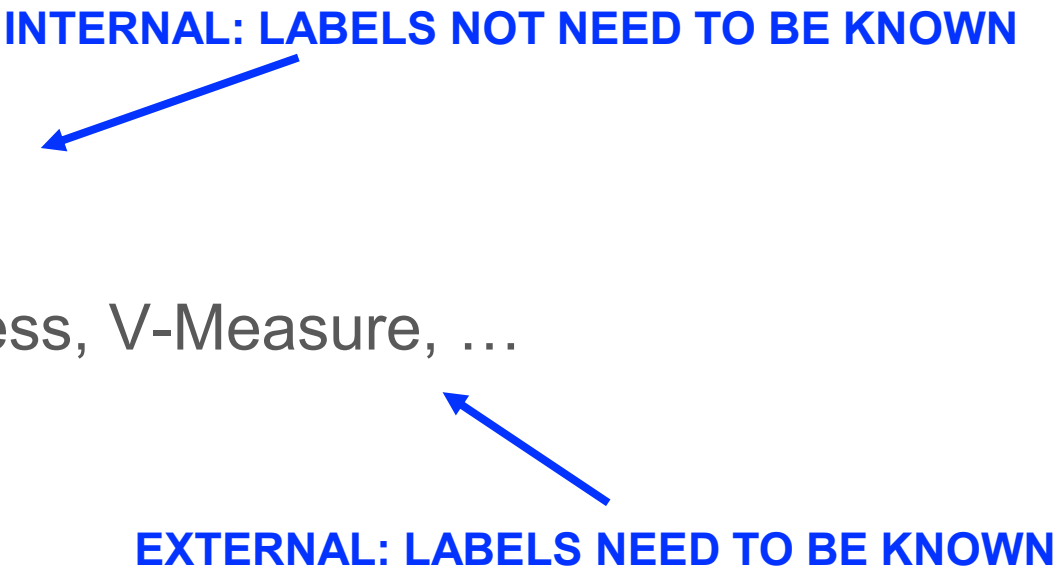
- Straightforward and **intuitive** algorithm

Cons

- Depends on **random initialization**
- Choose K **in advance**
- Clustering only by **linear separations**

Clustering

Clustering indices:

- Dunn Index, Silhouette, ...
 - Homogeneity, Completeness, V-Measure, ...
- INTERNAL: LABELS NOT NEED TO BE KNOWN
- EXTERNAL: LABELS NEED TO BE KNOWN
- 

Clustering

Dunn Index

$$DI = \frac{\min_{i,j|C(i)=C(j)} s(x_i, x_j)}{\max_{i,j|C(i) \neq C(j)} s(x_i, x_j)}$$

Minimum intra-cluster similarity

Maximum inter-cluster similarity

Silhouette

$$s = \frac{1}{n} \sum_{I=1}^n \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

-1 ≤ s ≤ 1

Average dissimilarity of i with points in the same cluster

Minimum average dissimilarity of i with points of other clusters

Clustering



Homogeneity (H)

- Each cluster contains only points of a single label

Completeness (C)

- All points of a label are assigned to the same cluster

V-Measure

- Harmonic mean between H and C

Clustering



Hierarchical clustering

- Start with **pairwise distances** between data points
- Iteratively **agglomerate** clusters
- Need to define a **distance between groups**
- At the end we obtain a **tree**, named **dendrogram**

Clustering

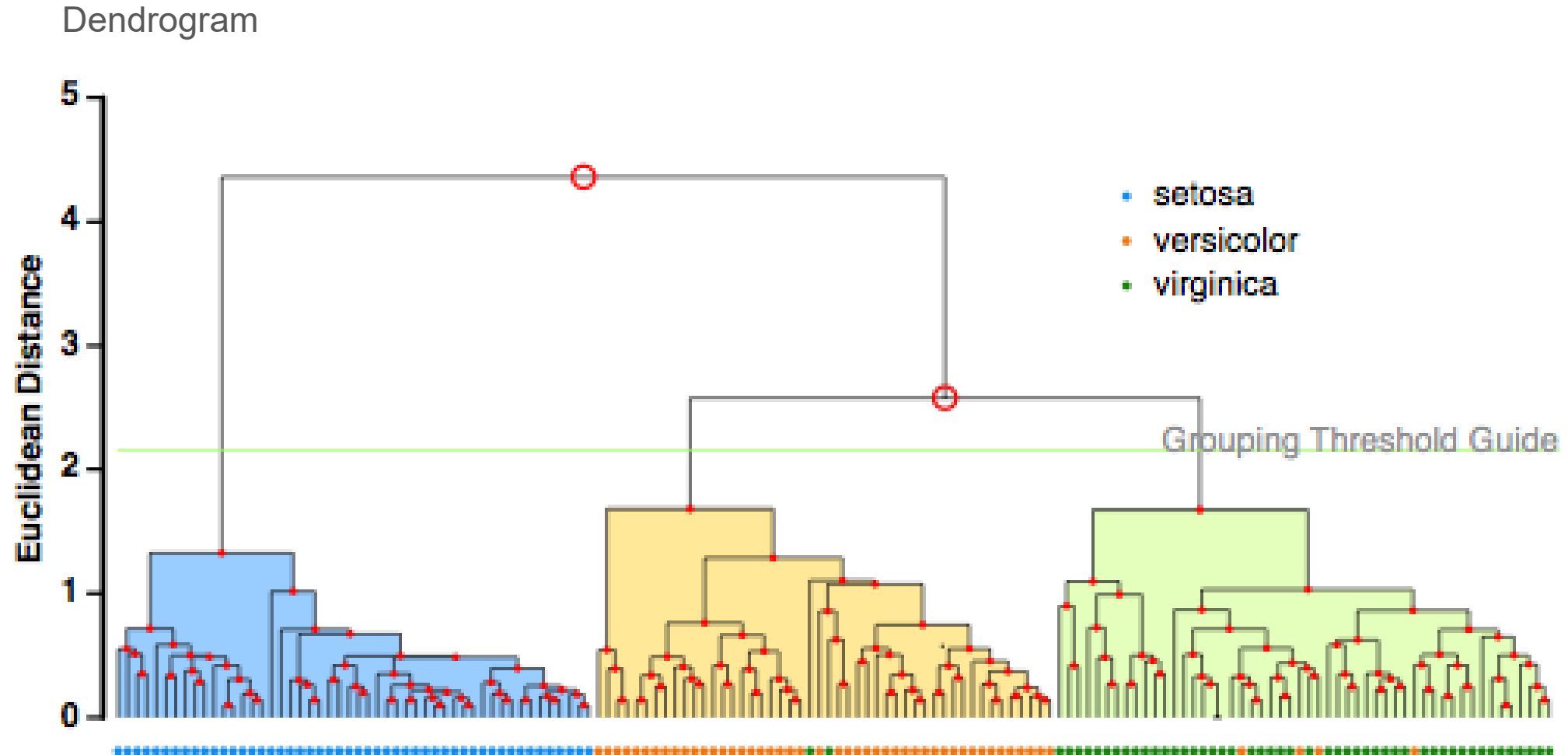
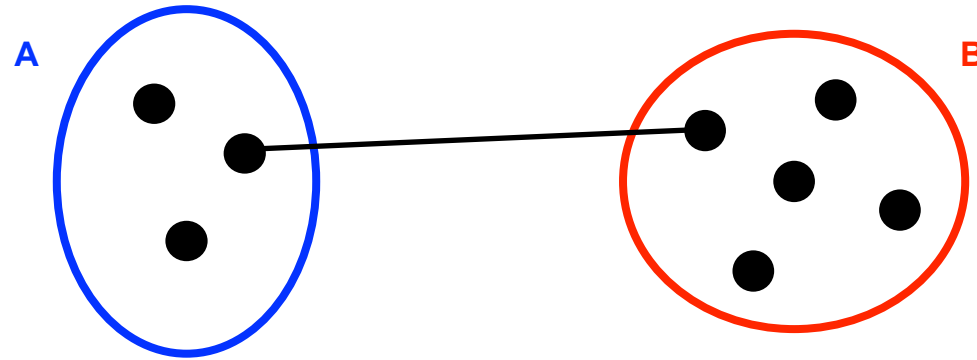


Figure from <https://www.gigawiz.com/hcluster2.html>

Clustering

Distances between groups of points:

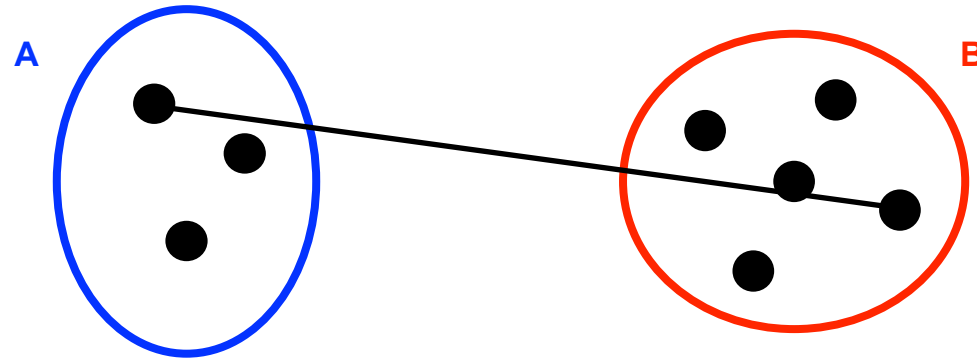


**SINGLE
LINKAGE**

$$d(A, B) = \min_{x \in A, y \in B} d(x, y)$$

Clustering

Distances between groups of points

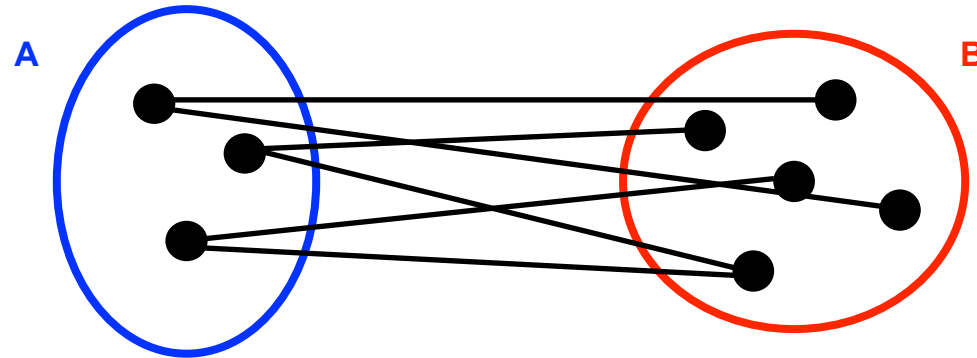


**COMPLETE
LINKAGE**

$$d(A, B) = \max_{x \in A, y \in B} d(x, y)$$

Clustering

Distances between groups of points

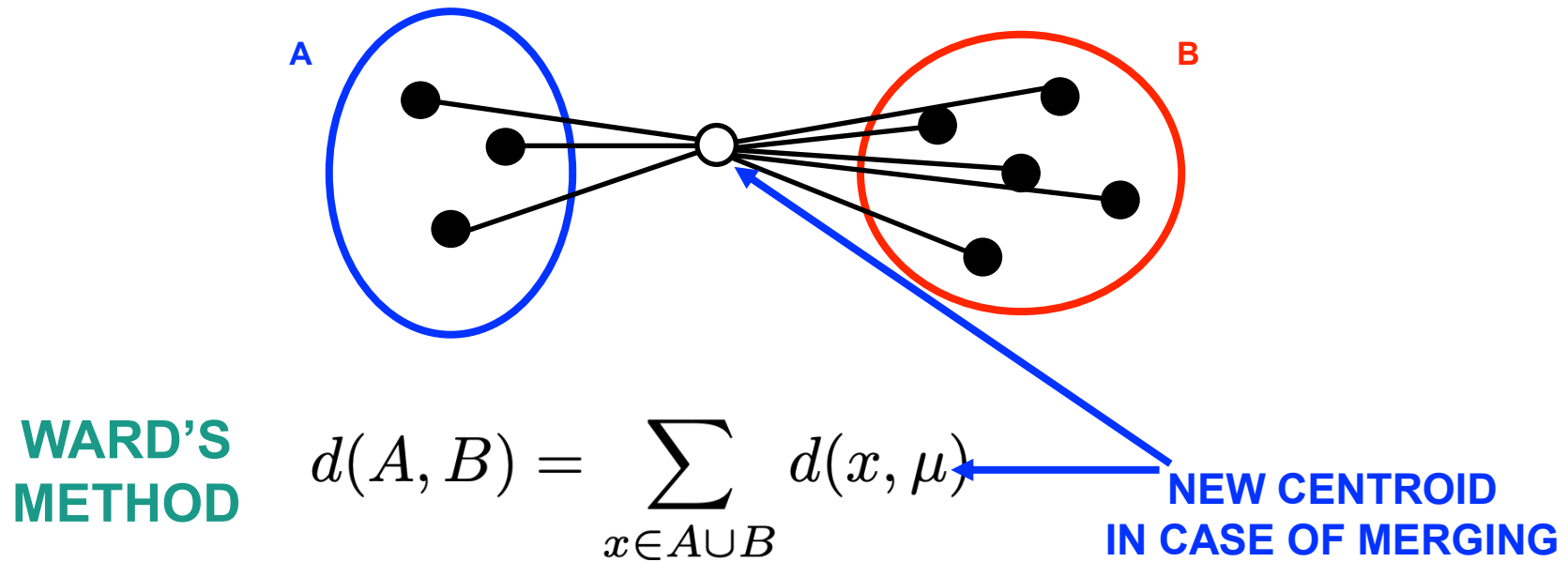


**AVERAGE
LINKAGE**

$$d(A, B) = \frac{1}{|A||B|} \sum_{x \in A, y \in B} d(x, y)$$

Clustering

Improvement in total distance in case of merging



Clustering



Many other algorithms...

- Spectral clustering
- Affinity propagation
- DBScan (it also estimates K)
- ...